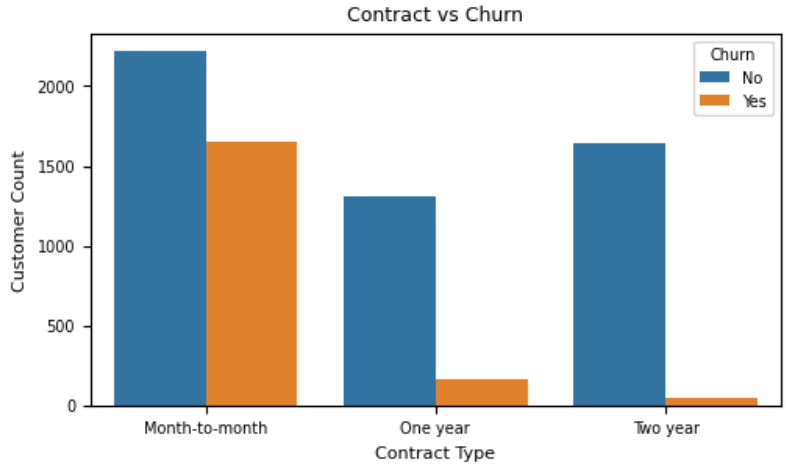
```
plot_df = pd.read_csv(filename)
plot_df["TotalCharges"] = pd.to_numeric(
    plot_df["TotalCharges"], errors="coerce"
)
plot_df = plot_df.drop_duplicates()

fig, ax = plt.subplots(figsize=(5, 3))

sns.countplot(
    data=plot_df,
    x="Contract",
    hue="Churn",
    ax=ax
)

ax.set_title("Contract vs Churn", fontsize=9)
ax.set_xlabel("Contract Type", fontsize=8)
ax.set_ylabel("Customer Count", fontsize=8)
ax.tick_params(axis="x", labelsize=7)
ax.tick_params(axis="y", labelsize=7)
ax.legend(title="Churn", fontsize=7, title_fontsize=7)

plt.tight_layout(pad=0.3)
plt.savefig(
    "Question8_CountPlot.png",
    dpi=100,
    bbox_inches="tight",
    pad_inches=0.05
)
plt.show()
```



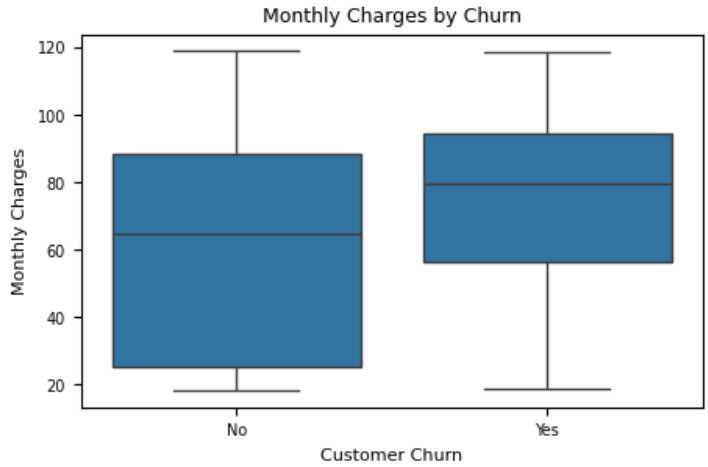

```
fig, ax = plt.subplots(figsize=(4.5, 3))

sns.boxplot(
    data=plot_df,
    x="Churn",
    y="MonthlyCharges",
    ax=ax
)

ax.set_title("Monthly Charges by Churn", fontsize=9)
ax.set_xlabel("Customer Churn", fontsize=8)
ax.set_ylabel("Monthly Charges", fontsize=8)
ax.tick_params(axis="both", labelsize=7)

plt.tight_layout(pad=0.3)
plt.savefig(
    "Question9_BoxPlot.png",
    dpi=100,
    bbox_inches="tight",
    pad_inches=0.05
```

```
)
plt.show()
```



10. Correlation Heatmap of All Numerical Features

```
numeric_df = df.select_dtypes(include=np.number)
correlation = numeric_df.corr()

fig, ax = plt.subplots(figsize=(6, 4))

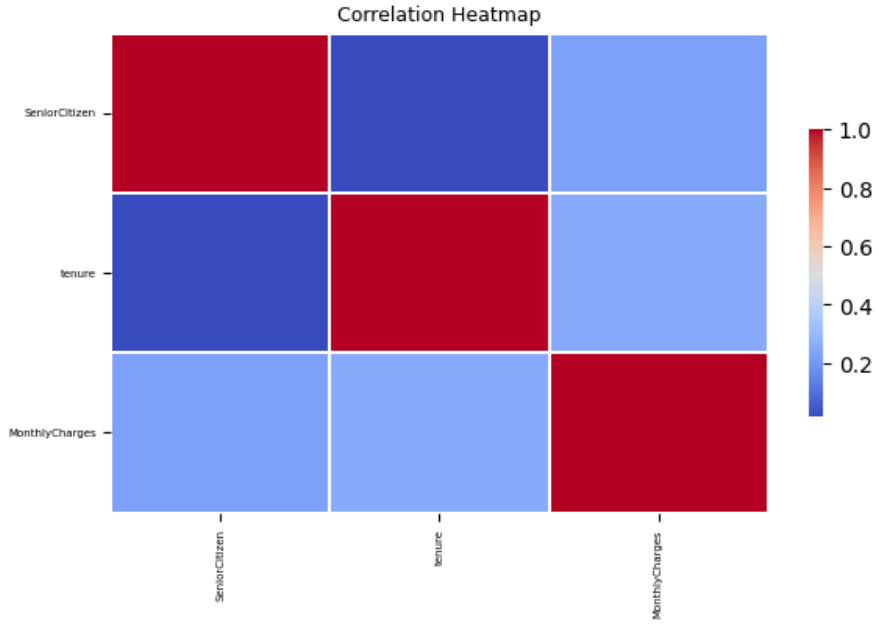
sns.heatmap(
    correlation,
    annot=False,
    cmap="coolwarm",
    linewidths=0.1,
    cbar_kws={"shrink": 0.6},
    ax=ax
)

ax.set_title("Correlation Heatmap", fontsize=9)
ax.tick_params(axis="x", labelsize=5, rotation=90)
ax.tick_params(axis="y", labelsize=5, rotation=0)

plt.tight_layout(pad=0.3)

plt.savefig(
    "Question10_Heatmap.png",
    dpi=100,
    bbox_inches="tight",
    pad_inches=0.05
)

plt.show()
```



Start coding or generate with AI.